

RGCS Phryll v2 — Build Sheet

Design PHV2-CRY-DEMO-120 · crystal CRY-DEMO-120

Crystal profile

crystal ID	CRY-DEMO-120
length (mm)	120
top / base diameter (mm)	26 / 39
Eye z (mm)	62.5

Generated parts

custom cone shell, coil sleeve with grooves, base adapter, cap, LED holder, jack holder, locker — all generated from the crystal envelope; no stock mesh scaled

Print settings

wall (mm)	1.8
clearance (mm)	0.66
print tolerance (mm)	0.2
suggested material	PLA / PETG

Coil settings

wire	AWG28 (0.33 mm)
copper	enameled copper — clockwise
silver	silver or non-ferrous source-selected conductor — counter_clockwise
turns per coil	60
winding band (mm)	32.5 – 92.5
electrical contact	none permitted between coils

Wire spacing

clear gap (mm)	0.66
groove pitch (mm)	0.99
groove depth (mm)	0.25

Coil-to-crystal standoff

nearest conductor (mm)	2.21
coil centerline (mm)	2.375

Eye alignment

Eye z (mm)	62.5
crossing plane z (mm)	62.5
alignment residual (mm)	0
tolerance (mm)	0.5
aligned	PASS

Pulse metadata (source-language, recorded not validated)

drive	two coils pulsed alternately at 4096 Hz
listed starting voltage	20 V (source-reported; user limits govern)
crossing	copper/silver crossed, no contact

Assembly notes

Wind copper clockwise and silver counter-clockwise in their grooves; verify the crossing plane against the Eye mark before fixing; check no electrical contact between coils with a continuity meter.

Claim boundary

This build sheet is an engineering plan and reproducibility record. Source-language pulse notes are recorded, not validated. Predictions are model outputs. Measurements decide.